

BREATHEBOX · REQUIREMENTS

BreatheBox requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/breathebox		
Source	docs/03-requirements.md @ 5c37130		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

BreatheBox requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design (see BBX-PRB-001). Status is judged against the first-order estimates in BBX-PRC-001; nothing has been built or measured.

The **design case** is a 30 m³ bedroom with two sleeping adults, 20 °C and 50 % relative humidity indoors, 0 °C outdoors, and a vertical sliding sash window with a 900 mm clear width.

Table 1. Requirements and status at TRL 2.

ID	Requirement	Target	Status at TRL 2	Verification (TRL 3 or later)
R1	Balanced ventilation over a useful range	Supply and exhaust each adjustable from 30 to 70 m ³ /h (18 to 41 cfm); supply and exhaust within 10 % of each other outside frost mode	At risk: fan operating point against about 80 Pa is unverified	Fan curve against a system pressure estimate; later flow hood measurement
R2	Recover heat from the exhaust air	Sensible effectiveness 75 % or more at 50 m ³ /h and balanced flow	Met on paper: about 80 % (estimate from core class)	Core data sheet; later temperature measurements on four ports
R3	Low running power	Fans and controls 12 W or less at 50 m ³ /h	Met on paper: about 9.5 W (estimate)	Fan data; later power meter
R4	Keep bedroom air fresh	CO ₂ at or below 1,000 ppm (24-hour average) in the design case, following the Health Canada residential guideline	Met at 70 m ³ /h (about 820 ppm, estimate); marginal at 50 m ³ /h (about 980 ppm, estimate)	Mass balance calculation; later CO ₂ logging
R5	Filter the supply air	ISO 16890 ePM1 50 % or better on supply; coarse filter on exhaust to protect the core	Met by specification	Filter data sheet
R6	Quiet at night	30 dB(A) or less at 1 m from the room face at 50 m ³ /h	Not verified, at risk	Fan noise data and duct attenuation estimate; later sound meter
R7	Handle condensate and frost	All condensate drains outdoors (up to about 0.15 L/h); runs without core blockage down to -10 °C outdoor, using a frost mode that may reduce supply flow	At risk: frost behavior below about -5 °C is unverified	Psychrometric calculation; later cold-chamber or winter test
R8	Fit common windows	Vertical sliding sash windows with 700 to 1,000 mm clear width and sill 150 mm or deeper; adapter panel for horizontal sliders	Partly met: sash and slider only; casement and tilt-and-turn windows not met	Model check against a window survey
R9	Renter-friendly installation	No drilling of frame or wall; one person installs or removes in 30 min or less; unit mass 12 kg or less	Met on paper: about 10 kg (estimate)	Fitting sequence review; later timed trials
R10	Electrical safety	Only 24 V SELV inside the unit, from a certified plug-in adapter; fused low-voltage input; no user-accessible mains	Met by design	Design review
R11	Weather and outdoor air	Rain-proof downward-facing hoods; insect mesh 1.5 mm or finer; 400 mm or more between supply intake and exhaust outlet	Met on paper (about 420 mm)	Model check; later spray test
R12	Security and child safety	Raised sash lockable onto the insert panel; no opening wider than 100 mm through the installed unit	Met by design, unverified	Design review

ID	Requirement	Target	Status at TRL 2	Verification (TRL 3 or later)
R13	Serviceable	Filters changed without tools in 2 min or less; core removable for washing; all parts replaceable with a screwdriver	Met by design	Design review
R14	Local data only	CO ₂ , humidity and temperature shown on the unit; works with no network or cloud account; optional local logging	Met by design	Firmware sketch review at TRL 3
R15	Affordable	Prototype parts \$220 or less	Not met: about \$243 (indicative)	Priced BOM (<code>bom/</code> <code>bom.csv</code>)

■ Requirements not met or at risk

- **R15 (cost) not met:** about \$243 against \$220. A budget change is proposed in `docs/REVIEW.md`, awaiting Amish.
- **R8 (fit) not met for casement and tilt-and-turn windows.** A separate insert is needed for those markets.
- **R1, R6 and R7 at risk:** the fan operating point, night-time noise and cold-weather frost behavior are unverified and are the main TRL 3 checks.
- **R4 marginal at 50 m³/h:** the design case needs the 70 m³/h boost while two people sleep, which makes R6 harder.

■ Assumptions

- Air density 1.2 kg/m³ and specific heat 1,005 J/(kg·K).
- CO₂ generation about 0.014 m³/h per sleeping adult (estimate); outdoor CO₂ about 420 ppm.
- The core behaves like a typical polymer plate counterflow core of this size; effectiveness falls as flow rises.
- System pressure at 50 m³/h about 80 Pa across filter, core, hoods and grilles (estimate).